ANSWER-1

Power BI is a business intelligence and data visualization tool used to analyze data and create interactive dashboards and reports. Businesses use Power BI because it helps them make data-driven decisions, combine data from multiple sources, automate reporting, collaborate easily, and gain real-time insights.

ANSWER-2

The three main components of Power BI are:

1. **Power BI Desktop** – A Windows application used to connect, clean, model, and visualize data to create reports.
2. **Power BI Service** – A cloud platform used to publish, share, and collaborate on reports and dashboards.
3. **Power BI Mobile** – Mobile apps that allow users to access and interact with Power BI reports from anywhere.

ANSWER-3

The Power BI workflow consists of three main steps:

1. **Data Loading –** Connecting to various data sources and importing data into Power BI.
2. **Data Transformation –** Cleaning and shaping the data in Power Query to make it analysis-ready.
3. **Data Visualization & Reporting –** Creating interactive reports and dashboards in Power BI Desktop and publishing them for sharing.

ANSWER-4

Four data cleaning tasks in Power Query are:

1. **Remove Duplicates** – Eliminates repeated rows to maintain data accuracy.
2. **Handle Missing Values** – Replace or remove null values to clean the dataset.
3. **Change Data Types** – Convert columns to correct data types for proper analysis.
4. **Split Columns –** Divide a column into multiple columns using delimiters (e.g., splitting names)

ANSWER-5

Steps to load the Real Estate dataset:

1. Open Power BI Desktop.
2. Click on **Get Data** from the Home tab.
3. Select the appropriate file type (CSV) and click **Connect**.
4. Browse and choose the Real Estate dataset file.
5. Preview the data and click **Load**.
6. Wait for the dataset to appear in the Fields pane.
7. Begin cleaning or creating visuals.

ANSWER-6

Power BI Desktop provides **three main views**, each serving a different purpose in the data analysis workflow. Understanding these views helps you clean data, build visuals, and manage relationships effectively.

1. Data View

Definition: Data View allows you to **see the actual data** that has been loaded into Power BI after it has been cleaned or transformed in Power Query.

Purpose:

- Inspect tables and columns
- Verify data types
- Check if transformations were applied correctly
- Perform simple calculations using DAX

This view is mainly used for **data validation and column-level analysis**.

2. Report View

Definition: Report View is where you **create visualizations, dashboards, and interactive reports**.

Purpose:

- Build charts, graphs, maps, and KPIs

- Add slicers and filters
- Arrange visuals into report pages
- Design the final layout for presentation

This is the primary workspace for **data visualization and storytelling**.

3.Model View

Definition: Model View displays the **relationships between tables** in your dataset.

Purpose:

- Create and edit relationships
- View cardinality (one-to-many, many-to-one)
- Manage table connections
- Organize data models for better performance

This view is essential for **data modeling and ensuring tables interact correctly**.

ANSWER-7

Power BI supports a wide range of data sources, including

1. File-Based Data Sources

Power BI can import data from common file formats such as:

- Excel (.xlsx)
- CSV (.csv)

2. Database Sources

Power BI connects directly to many databases, including:

- SQL Server
- MySQL

3. Online Services

Power BI supports cloud-based services such as:

- SharePoint
- OneDrive

4. Web Data Sources

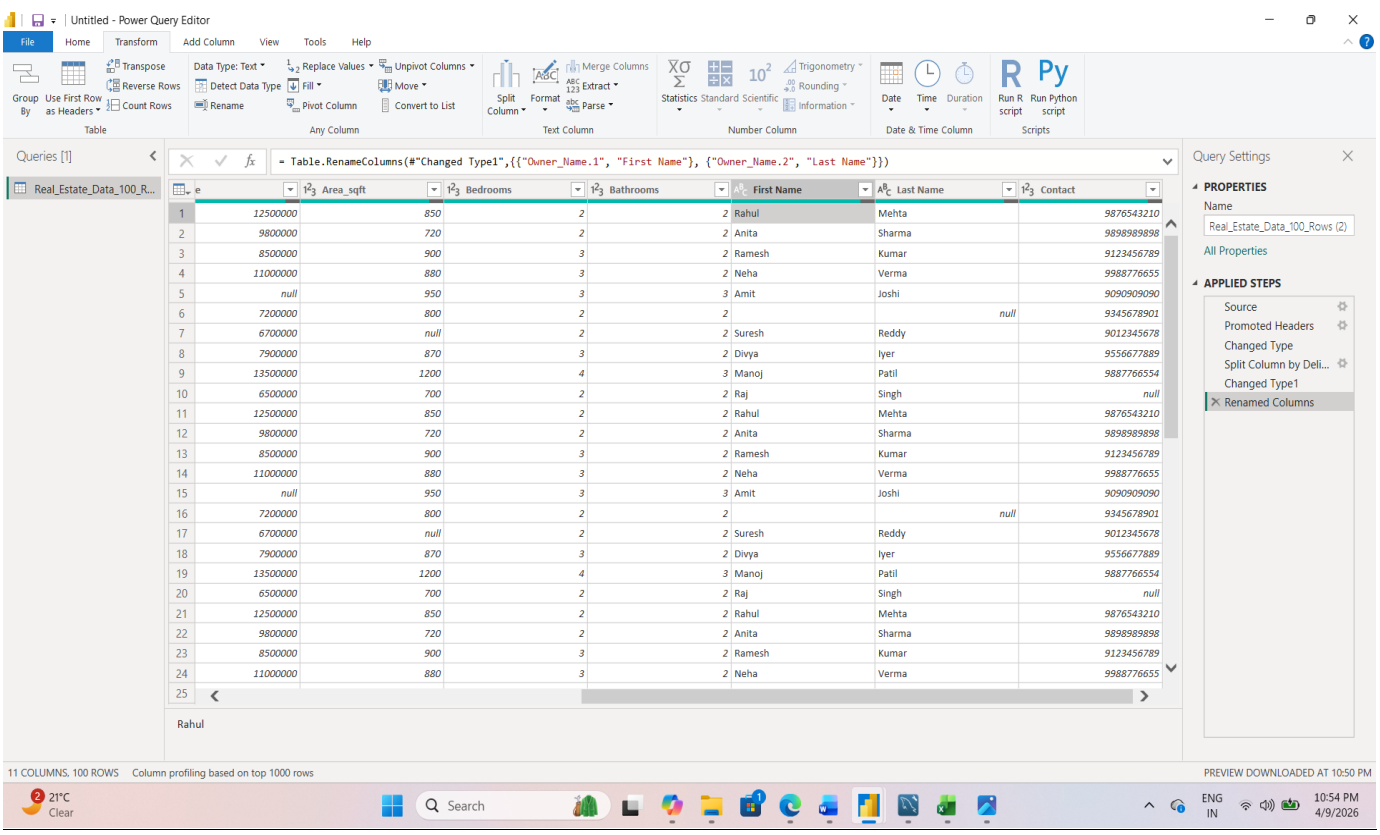
You can load data from:

- Websites
- Web APIs

5. Other Sources

Power BI also supports:

- Power BI datasets
- Power BI dataflows
- OData feeds

ANSWER-8

"Using Power Query, I selected the 'Owner Name' column, used Split Column → By Delimiter with space as the delimiter, which created two columns. I then renamed them to 'First Name' and 'Last Name' and applied the changes."